COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE0010 / DATA STRUCTURE ALGORITHMS	**SECTION:**	CPE21S4
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SEATWORK 7.1 USING SORTING ALGORITHMS

1. PROCEDURE OUTPUT

1. What is a sorting algorithm?
   - A sorting algorithm is a method used to arrange data in a specific order. The data can usually be arranged from smallest to largest or largest to smallest.

2. Where can sorting algorithms be used?
   - Sorting algorithms can be used in many programs and systems where data needs to be organized.
   - Arranging student grades from highest to lowest
   - Organizing names alphabetically
   - Sorting products according to their prices
   - Arranging files by name or date
   - Organizing search results
   - Sorting scores in games
   - Arranging records in databases

3. Explain the different types of sorting algorithms.

Bubble Sort - works by comparing two numbers beside each other. If they are in the wrong order, they are swapped. This process continues until the numbers are properly arranged

Selection Sort - works by finding the smallest value from the unsorted part of the list and placing it in the correct position. It repeats this process until all the values are sorted.

Insertion Sort - works somewhat like arranging cards in your hand. It takes one value at a time and places it in its proper position among the values that have already been sorted.

4. Sample C++ programs using sorting algorithms

Selection Sort

```
#include <iostream>
using namespace std;


int main() {
```

```
    int arr[] = {12, 11, 13, 5, 6};
    int n = 5;

    for (int i = 1; i < n; i++) {
        int key = arr[i];
        int j = i - 1;

        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j--;
        }

        arr[j + 1] = key;
    }

    cout << "Sorted array: ";

    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }

    return 0;
}
```

The program starts by assuming that the first number is the smallest. It then checks the remaining numbers to find if there is a smaller value. Once the smallest number is found, it is exchanged with the number in the current position. The process repeats until the entire array is sorted.

Insertion Sort

```
#include <iostream>

using namespace std;
```

```
int main() {
    int arr[] = {12, 11, 13, 5, 6};
    int n = 5;

    for (int i = 1; i < n; i++) {
        int key = arr[i];
        int j = i - 1;

        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = key;
    }
    cout << "Sorted array: ";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

The program starts with the second element and compares it with the numbers before it. If the previous number is bigger, it moves that number one position to the right. The selected number is then placed in its correct position. This continues until all the numbers are arranged.

Bubble Sort

```
#include <iostream>

using namespace std;


int main() {
    int arr[] = {5, 1, 4, 2, 8};
    int n = 5;


    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }


    cout << "Sorted array: ";


    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

The program compares two values that are next to each other. If the first value is greater than the second value, they are swapped. After every round, the larger values gradually move toward the end of the array. The program continues doing this until the values are in ascending order.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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